Sadeep Ariyarathna

Mechanical Desing Engineer | Mechatronics Engineer

Personal Information

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Experience

Intern / Masterstudent – E/E Architecture
CLAAS E-Systems GmbH, Dissen am Teutoburger Wald, Germany
Oct 2024 – Aug 2025

- Developed and implemented an **automated CI/CD workflow** using **Azure DevOps** and **LemonTree** to integrate system architecture models more efficiently, reducing merge time by ~75%.
- Created a KPI-based evaluation framework to assess workflow performance using user feedback, time trials, and success rates.
- Contributed to improving system usability (**Enterprise Arhcitects, Lemontree and Azure DevOps**) and process transparency within the E/E architecture development team.

Research Assistant
Fachhochschule Südwestfalen, Soest, Germany
Mar 2024 – Aug 2024

- Engineered a **C-based** application on an STM32 microcontroller (F446RE) to collect real-time data from an MCP9600 temperature sensor via **I2C** and transmit it over a cellular network using a SIM7600 GSM module.
- Streamlined data processing and transmission by leveraging Direct Memory Access (DMA) and AT commands to efficiently publish JSON-formatted data payloads to an **MQTT** broker.

Mechatronics Engineer
AtlasLabs (Pvt.) Ltd, Sri Lanka
Sep 2022 – Feb 2023

- Engineered and tested mechanical components and systems from concept to launch using **Fusion 360**, reducing product development cycle time by 20%.
- Created manufacturable designs and validated them according to **Design for Manufacturing** (DfM) and **Design for Assembly** (DfA) principles.
- Reduced troubleshooting time by 35% by introducing standardized **bills of materials and validation documentation**.

Engineering Intern
MAS Intimates (Pvt.) Ltd, Sri Lanka
Jul 2021 – Feb 2022

- Designed and developed **SolidWorks**-based attachments for sewing machines, reducing material cutting time by 65%.
- Created mechanical control systems using **Arduino** and **PLC** (Delta) to evaluate multiple conceptual prototypes.
- Supported cross-functional teams in implementing manufacturing process improvements.

Education

Master of Science in Systems Engineering and Engineering Management
Fachhochschule Südwestfalen, Soest, Germany
Apr 2023 – Sept 2025

- Thesis: *A Collaboration Workflow for Merging System Architecture Models – An Example from the Agriculture Industry (Marks - 1.0)*
- Final GPA – 1.6 (German grading system)

Bachelor of Science in Mechatronics
General Sir John Kotelawala Defence University, Sri Lanka
Jan 2017 – Dec 2021

- Thesis: *Development of Elbow Prosthetic Control System*
- *Final GPA – 3.59*

Projects

Smart Platform
A self-stabilizing platform build using **C++**. System contains a motion sensor, **Arduino** Uno, a motor driver driving a stepper motor controlled by **PID**. All the components are included in a **custom PCB**.

Remote Motion Detection
An attachment developed using a **Raspberry Pi pico W** and a motion sensor to detect rotations and displacements around the 3 axis and transfer the data through **MQTT** to a **NodeRed** dashboard.

Gesture Control System
Created a computer vision program using **OpenCV** and **Python** to recognize specific finger gestures for controlling system functions such as opening applications or typing commands.

Zone Temperature Control
A master temperature control system simulation for individual zones with time-based scheduling capabilities, designed with **CODESYS** using **Structured Text** (ST) and an **HMI** design to visualize and control states.

Technical Skills

Programming	Python, C/C++, MATLAB, Arduino, PLC (Delta), CODESYS
Modeling & Design	SolidWorks, Fusion 360, Solid Edge, Ansys, MATLAB Simulink, STM32 CubeIDE, KiCAD
Developer Tools	Git, Azure DevOps, Docker, MS Project, Jira
Libraries & Frameworks	PyTorch, TensorFlow, Stable Baselines, OpenAI Gym, OpenCV
Other Tools	Microsoft Office, Adobe Photoshop, HTML & CSS

Languages

English	C1 (Fluent - IELTS)
German	A2 (Limited Proficiency)

Additional Skills

- Hands-on experience with **workshop tools, soldering,** and **PCB design**
- Strong analytical and problem-solving mindset